

PRACTICE TEST

For questions 1 and 2, choose the correct answer: A, B, C, or D

1. For the number 64, which statement is not true?
 - A. It has only one factor.
 - B. It is a perfect square.
 - C. It is a perfect cube.
 - D. Its prime factor is 2.
2. The factorization of the trinomial $2x^2 + 7x + 6$ is:
 - A. $(2x + 1)(x + 6)$
 - B. $(2x + 2)(x + 3)$
 - C. $(2x + 3)(x + 2)$
 - D. $(2x + 6)(x + 1)$
3. Write each number below as a product of its prime factors, then determine the least common multiple and the greatest common factor of the 3 numbers.
20 45 50
4. Look at the numbers in question 3.
 - a) What would you have to multiply each number by to produce a number that is:
 - i) a perfect square?
 - ii) a perfect cube?
 - b) Why is there more than one answer for each of part a?
5. Expand and simplify, using the strategy indicated.
Sketch a diagram to illustrate each strategy.
 - a) Use algebra tiles: $(2c + 5)(3c + 2)$
 - b) Use an area model: $(9 + 4r)(8 + 6r)$
 - c) Use a rectangle diagram: $(4t - 5)(3t + 7)$
6. Expand and simplify.
 - a) $(2p - 1)(p^2 + 2p - 7)$
 - b) $(e + 2f)(2f^2 + 5f + 3e^2)$
 - c) $(3y + 2z)(y + 4z) - (5y - 3z)(2y - 8z)$
7. Factor each polynomial. For which trinomials could you use algebra tiles? Explain.
 - a) $f^2 + 17f + 16$
 - b) $c^2 - 13c + 22$
 - c) $4t^2 + 9t - 28$
 - d) $4r^2 + 20rs + 25s^2$
 - e) $6x^2 - 17xy + 5y^2$
 - f) $h^2 - 25j^2$
8. A cube has edge length $2r + 1$. A right square prism with dimensions r by r by $2r + 1$ is removed from the cube.
Write an expression for the volume that remains. Simplify the expression.
9. Write all the trinomials that begin with $8t^2$, end with $+3$, and can be factored.
How do you know you have found all the trinomials?